TED (21) -5283A **REVISION 2021**

Model Question Paper

Data and Application Security

Time : 3 Hour ***Max.Marks : 75***

PART A

I. **Answer all the following questions** **(9 x 1 = 9 Marks)**

1	What is cryptography?	M1.01	R
2	What is a Vigenère cipher?	M1.02	U
3	What are the two types of stream ciphers?	M2.03	R
4	What does PKCS stand for?	M2.01	R
5	What is Elliptic Curve Cryptography (ECC)?	M2.02	U
6	What is pseudo-randomness?	M3.03	R
7	Which secret sharing scheme is based on modular arithmetic?	M3.02	U
8	How can HTTP cookies be used for secure web application communication?	M4.02	U
9	What are hidden form fields used in web application security?	M4.02	U

PART B

II Answer any 8 questions from the following .Each questions carry 3 marks.

(8 x 3 = 24 Marks)

1	What is probability theory in cryptography?	M1.01	R
2	What is a symmetric (private) key cryptographic system?	M1.02	U
3	Describe the Merkle-Hellman cryptosystem and discuss its security considerations.	M2.02	U
4	Explain the purpose of non-linear feedback shift registers (NLFSRs) in stream ciphers.	M2.03	U
5	Discuss the security of the RSA cryptosystem and the role of primality testing in RSA key generation.	M2.02	U
6	Explain the properties of a digital signature and why they are important in ensuring the security of digital communications.	M3.01	U
7	Define threshold secret sharing and explain its significance in securing sensitive information.	M3.02	U
8	Explain the concept of statistical tests of randomness and provide examples of commonly used tests.	M3.03	R
9	What are common vulnerabilities in access controls?	M4.03	R
10	Explain Authentication and authorization	M4.02	R

PART C

III Answer ALL questions. Each carry 7 marks. (6 x 7 = 42 Marks)

1	Discuss the characteristics and working principles of symmetric key cryptographic systems, including Caesar, Affine, Monoalphabetic Substitution, Transposition, Homophonic Substitution, Vigenère, Beaufort, and the DES family **OR**	M1.02	Analysing
2	Symmetric Key Cryptographic Systems: a) Encrypt the message "HELLO" using a Caesar cipher with a shift of’3 b) Apply the Vigenère cipher to the plaintext "CRYPTO" using the keyword "KEY" for encryption.	M1.03	Applying
3	Compare and contrast the RSA cryptosystem and the ElGamal cryptosystem in terms of their key generation, encryption, and decryption processes **OR**	M2.01	Analysing
4	Discuss the significance of feedback shift registers (FSRs) in the design of stream ciphers. Explain the concepts of linear complexity and non-linear feedback shift registers (NLFSRs), and discuss their role in achieving higher security in stream cipher designs. Provide examples of stream ciphers that utilize FSRs or NLFSRs in their construction.	M2.03	U
5	Compare and contrast synchronous stream ciphers and self-synchronizing stream ciphers, discussing their advantages and limitations. **OR**	M2.03	Analysing
6	Explain the concept of PKCS (Public Key Cryptography Standards) and discuss one specific standard from the PKCS series in detail, highlighting its significance in asymmetric key cryptographic systems.	M2.01	U
7	Discuss the Blakley secret sharing scheme and its approach to distributing secret information among multiple participants. **OR**	M3.02	U
8	Compare and contrast the Rabin-Lamport signature scheme, the Matyasmeyer signature scheme, and the RSA signature scheme. Discuss their strengths, weaknesses, and potential applications.	M3.01	Analysing

9	Analyze the concept of randomness and pseudo-randomness, including the challenges in generating true random numbers and the need for pseudo-random number generators (PRNGs) in cryptography.	M3.03	Analysing
	OR		
10	Evaluate the advantages and disadvantages of using multiple RSA signatures over a single RSA signature. Consider factors such as performance, security, and key management.	M3.01	U
11	Discuss core defense mechanisms used in web application security and their significance.	M4.01	R
	OR		
12	Explain the concept of web spidering in the context of handling attackers in web application security	M4.01	R